

issue considering modern video processing hardware does a pretty good job at FRC/dithering to achieve true colour output despite the 6-bit panel limitation. H-IPS and S-IPS panels with true 8-bit panels are usually necessary in prohibitively expensive colour accurate monitors for professionals.

VA Panels

Vertical Alignment (VA) panels lie somewhere in the middle of the cost/quality spectrum of LCD displays. This panel type is further subdivided into S-PVA/MVA and offer significantly better colour as well as generous viewing angles when compared to TN panels. Again, like the IPS panels, VA panels tend to have slower response times. Since this technology is more or less similar to S-IPS, it also faces the same problems. Even though VA panels deliver great colour performance and wide viewing angles, it's still not as good as IPS panels. To make matters worse, the response times is worse when compared to IPS. These panels are definitely not the right fit for gamers, because they tend to add palpable input lag as well.

On the flip side, VA panels tend to offer better blacks even when compared to IPS due to their inherently better contrast ratios. Unfortunately, the performance isn't as balanced because these panels are marred by serious colour shift issues, which also lead to annoying brightness levels across the width, especially when viewing even from a slight off centre angle. What's more, even the central viewing position isn't immune to issues. Colour shifts tend to cause reduced darker details in dark scenes, even when viewed directly from the centre. While some users aren't sensitive to these colour shifts and uneven brightness, but still others deem it a deal-breaking issue. Thankfully the proliferation of cost effective IPS panels means that VA technology has more or less been edged out. You're better off avoiding monitors with these panels, since you're likely to find a good IPS alternative at roughly the same price range.

The World Beyond LCD

LED-backlit LCDs is doing great in the computer monitor space, but once you enter the realm of larger screens, it opens up the market for OLED alternatives. The sublime plasma technology may be dead, but its throne is taken over by another capable emissive alternative that is OLED, which fills in the gaps left behind by LED displays. What follows is an overview